

Applied Data Science 2025-2026

Teacher: Eugene Cruz | ecruz@ransomeverglades.org

Classroom: STEM 319 (Period 7) or STEM 304 (Period 6)

Office: STEM 205

Course Information

Description: This course introduces students to the emerging field of data science, an interdisciplinary field that uses the scientific method and algorithms to extract knowledge, trends, and patterns from data. In this course, students learn applications of statistics, information visualization, text analysis, machine learning, and social network analysis. The course starts with a few weeks of introductory Python for data science. It then continues with an extensive sequence of units in data analysis. During this part of the course, students work with Python toolkits such as pandas and data visualization libraries to manipulate, clean, and visualize information from various datasets. By the end of the first semester, students will work on data analysis projects using real-world data. In the second semester, students start an introductory course in machine learning using the scikit-learn library. Students participate in multiple machine learning projects, competitions, and challenges during the second semester as part of a hands-on experiential learning process, culminating in a semester-long capstone project on a topic of their choice.

Course Outline

This outline is subject to change. Each unit will usually have a MiniQuiz associated with it.

Semester 1: Introduction to Data Science.

- Unit 1: Introduction to Data Science Principles
- Unit 2: Data Analysis with Python
- Unit 3: Applied Plotting, Charting & Data Representation in Python
- Unit 4: Storytelling with Data: Concepts of Data Visualization
- Unit 5: Interactive Python Dashboards

Semester 2: Capstone Project.

- Unit 6: Applied Machine Learning in Python
- Unit 7: RE Project (Students will be working on an interdisciplinary project on RE)
- Unit 8: Kaggle Competitions
- Unit 9: Final Project

Texts/Supplementary Course Materials

Our course materials are fully online and free (though there are options to download and print materials).

Texts:

- Python Data Science Handbook. Jake VanderPlas
- Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems. Geron
- Introduction to Machine Learning with Python: A Guide for Data Scientists
- Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython
- Pandas for Everyone: Python Data Analysis (Addison-Wesley Data & Analytics Series)
- Storytelling with Data: A Data Visualization Guide for Business Professionals
- The Truthful Art: Data, Charts, and Maps for Communication

Web Resources:

- <https://github.com>
- <https://stackoverflow.com>
- <https://www.kaggle.com>
- <https://www.data.gov>
- <https://www.springboard.com/blog/free-public-data-sets-data-science-project/>

Video Channels:

- <https://www.youtube.com/channel/UCnVzApLJE2ljPZSeQylSEyg>
- <https://www.youtube.com/channel/UC68KSmHePPePCjW4v57VPQg>
- <https://www.youtube.com/channel/UC6ogRMJRjTuTskBnl-LkPAg>

Grade Breakdown

The course will be graded on a **points-based system**.

- Projects (both Individual and Group) are worth **100 points each**.
- In-class work, as well as homework, mini-quizzes, and participation grades, will be assigned between **1 and 20 points each**, depending on their scope.

Classroom Policies

- **Extra Help:** I am available most mornings (except Thursdays and Fridays), during unscheduled periods (3, 5, or 8), and after school until 4:00 p.m. Please schedule a time if needed. If I am unavailable, you are encouraged to consult with other teachers in the department.

- **Assignments and Assessments:** All work and due dates will be posted on MyCOMPASS. Major assessments will be posted at least 7 days in advance; smaller tasks at least 48 hours in advance. Planned absences require prior communication; unexcused absences follow late work penalties. Illnesses typically require completing work upon return unless other arrangements are approved.
- **Late Work:** Extensions may be requested in advance. Without an extension, late work incurs a 10% penalty per day. The minimum grade for completed late work is 50%; a zero is given only if no evidence of learning is submitted before the class moves on. Major assignments must be completed to earn credit for the course.
- **Academic Integrity:** Students must uphold the Honor Code: no lying, cheating, plagiarism, or creating the perception of impropriety. Unauthorized use of AI tools is prohibited. AI use will be clearly defined per assignment using Red (no AI), Yellow (guided/cited AI use), and Green (open AI use with accountability). All AI use must be disclosed and cited.

I have read and understood all of the above information:

Student Signature Date

Parent/Guardian Signature Date